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Section: B₁



Subject : Microbiology + Biochemistry

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Sr.No List of Microbiology Experiments:

- 1- Sterilization of glasswares
- 2- Simple staining technique of yeast
- 3- Gram staining technique of given culture
- 4- Preparation of nutrient agar plate.
- 5- Determination of milk quality
- 6- Microbiology of soil
- 7- Microbiology of air.



Experiment No : 1

★ Sterilization of glassware:-

Apparatus:-

- Test tubes
- Conical flask
- Petri disk
- Thread
- Pipette
- Wrapping paper
- Cotton
- Autoclave



Experiment No : 1

* Sterilization of glasswares :

Sterilization :-

It is the process of killing or removing bacteria or other microorganisms.

Principle :-

Killing of micro-organisms with hot air or steam by denaturing and coagulating its protein.

Apparatus :-

- i- Test tubes
- ii- Conical flask
- iii- Petri dish
- iv- Thread
- v- Pipettes
- vi- Wrapping paper
- vii- Cotton
- viii- Autoclave



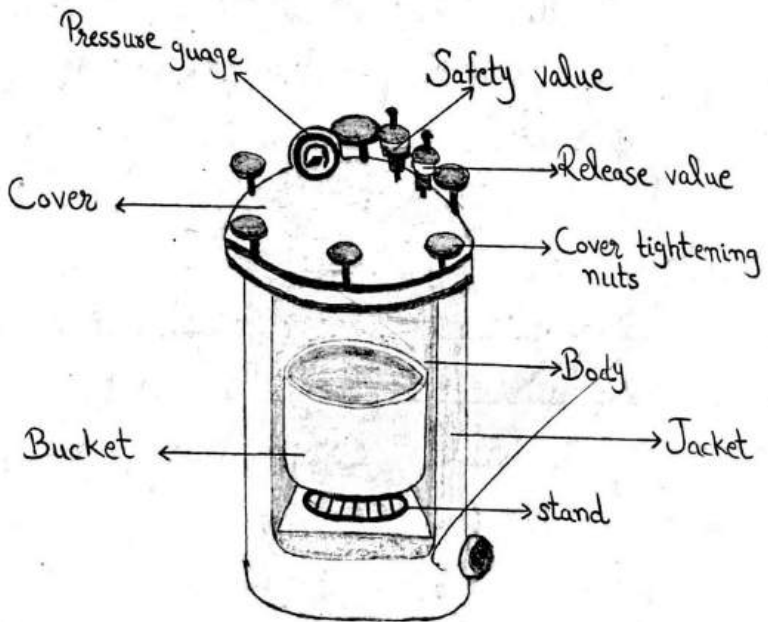


Fig: Autoclave

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Procedure:-

- i- A sufficient quantity of water is added to the base till it reaches bottom of perforated chamber.
- ii- The article to be sterilized is placed on perforated bottom by wrapping with the wrapping paper.
- iii- Now the lid is closed and set the temperature at 120°C for 15-20 min and on the autoclave.
- iv- As the temperature of water rises. It starts to vaporize and results in the form of steam.
- v- When pressure reaches the desired value. Keep it for corresponding holding time.
- vi- After this, switch off the autoclave and allow it to cool down.
- vii- After cooling, open the lid and remove the article.

Results:-

All the glass wares are

sterilized.

Precautions:-

- Glass wares should be washed properly.
- Lid of the autoclave should be opened carefully.

_____ Save



Experiment No: 2

★ Identification of bacterial Sample by using Gram Staining Technique:

Apparatus:-

- Glass slide
- Inoculating loop
- Glass pipette
- Bunsen burner
- Filter paper
- Microscope

Chemicals:-

- Crystal violet
- Iodine solution
- Safranin
- Acetone
- Water
- Bacterial sample



Experiment No: 2

* Identification of bacterial sample by using Gram staining technique:

Gram staining :-

It is a staining method used to classify bacterial species into two groups i.e.:

- i- Gram Positive bacteria
- ii- Gram Negative bacteria

Apparatus :-

- Glass slide
- Inoculating loop
- Glass pipette
- Bunsen burner
- Filter paper
- Microscope



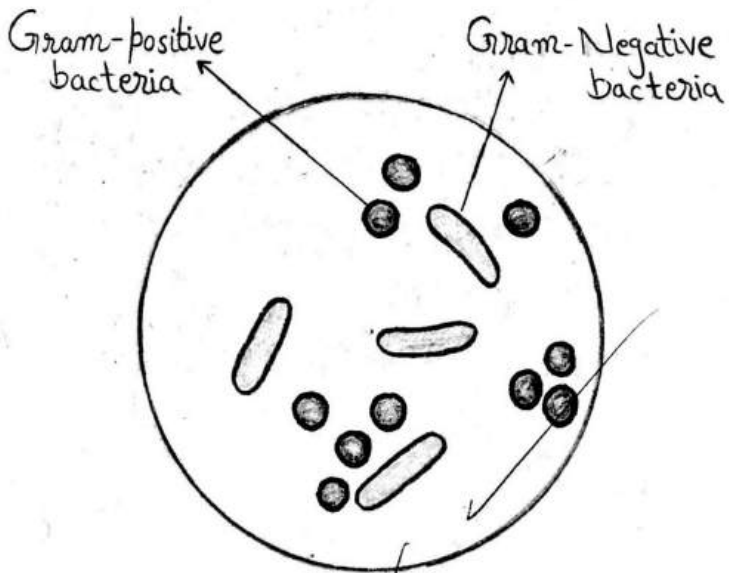


Fig: Gram staining technique



Chemicals:-

- Crystal violet
- Iodine solution
- Safranin
- Acetone
- Water
- Bacterial sample

Procedure:-

- i- Make a thin smear of material on glass slide by using pipette and dry it at room temperature.
- ii- Pass the slide through a flame so, that smear is fixed on it.
- iii- Now dip the glass slide with crystal violet dye for one minute.
- iv- Wash off the dye by placing the glass slide in iodine solution for one minute.
- v- Rinse in water
- vi- Then pour on acetone/alcohol until no more violet colour comes from the slide.

- vii- Rinse with water again
- viii- Again stain the slide with safranin dye.
- ix- Rinse with water
- x- Allow the glass slide to dry by using filter paper.
- xi- Finally, observe the slide under the microscope.

Results:-

Gram positive bacteria have thick cell wall so, they retain primary color and appears as purple in colour while Gram negative bacteria have thin cell wall so, they don't retain primary color and appear pink in color.

Precautions:-

- i- Smear should be accurately fixed.
- ii- Do not take excessive stain and water.

Experiment No: 3

* Simple staining of Yeast

Apparatus:-

- Glass slide
- Pipette
- Spatula
- Burner
- Filter paper
- Microscope

Chemicals:-

- Yeast sample
- Crystal violet dye
- Water

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Experiment No : 3

* Simple staining of yeasts

Yeast:-

Yeast are unicellular fungi and are larger than bacteria.

Apparatus:-

- Glass slide
- Pipette
- Spatula
- Burner
- Filter paper
- Microscope

Chemicals:-

- Yeast sample
- Crystal violet dye
- Water

Procedure:-

- i- Prepare the yeast sample by adding yeast granules in the water and mix it well, with the help of spatula.

Single yeast
cell

Budding yeast
cell

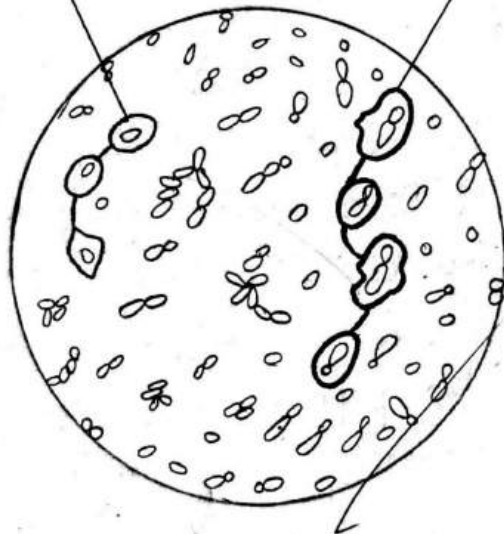


Fig: Staining of yeast



- ii- Pour the sample on the glass slide by using pipette and cover it with the cover slip.
- iii- Dry the glass slide properly on burner.
- iv- Then dip the glass slide in crystal violet dye.
- v- Wash the slide with water.
- vi- Dry the slide by using filter paper.
- vii- Observe the glass slide under microscope.

Result:-

Yeast cell stained with crystal violet are observed under microscope.

Precautions:-

- Smear should be fixed carefully.
- Do not take excessive stain.

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Experiment No : 4



Preparation of Nutrient Agar media for cultivation of bacteria :

Apparatus:-

- Conical flask
- Petri dish
- Glass rod
- pH meter
- Oven

Chemicals:-

- Beef extract
- Peptone
- Agar
- NaOH
- HCl
- Water

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Experiment No: 4

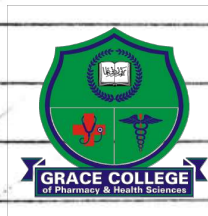
Preparation of nutrient agar media for cultivation of bacteria:-

Nutrient Agar:-

Nutrient agar media is a growth media for the cultivation of the bacteria in the laboratory.

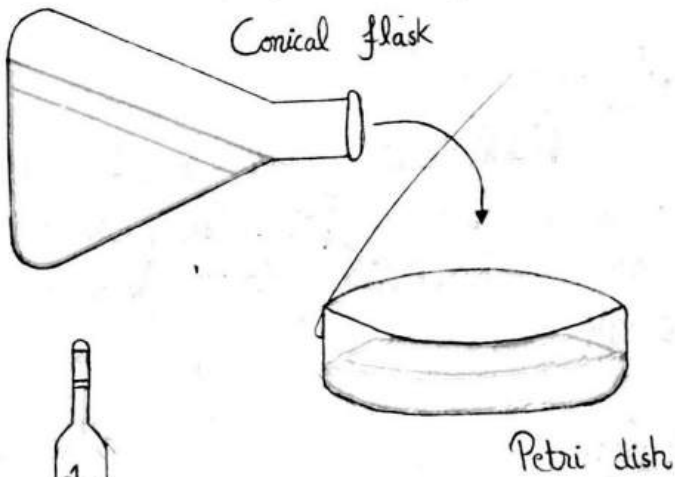
Chemicals:-

- Beef extract
- Peptone
- Agar
- NaOH
- HCl
- Water



Apparatus:-

- Conical flask
- Petri dish
- Glass rod
- PH meter



Pipette



• Oven / Incubator.

Procedure:-

- i- Take 70ml of distilled water in a conical flask.
- ii- Add 0.3g beef extract and 0.5g peptone in it and mix it properly.
- iii- Adjust the pH of solution within the range of by adding HCl or NaOH accordingly and make volume of solution 100ml by adding water.
- iv- Now add 1g agar in the conical flask containing solution and heat it in a oven for 1-2 minutes.
- v- Autoclave the conical flask containing nutrient agar media.
- vi- Pour the nutrient agar media in a petri dish and allow to solidify.

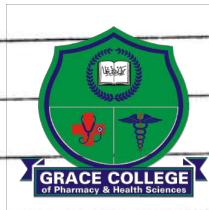
Result:-

A pale yellow culture media

was prepared.

Precautions:-

- i- All the apparatus were washed properly before use.
 - ii- Glass wares were handled properly.
 - iii- Chemicals were measured accurately.
 - iv- pH of meter was maintained properly.
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Experiment No : 5

★ Determine the quality of milk
by methylene blue dye
reductase test :

Apparatus :

- Test tube
- Test tube stand
- Glass rod
- Pipette

Chemicals :

- Methylene ^{blue} dye
- Milk sample



Experiment No : 5

Determine the quality of milk
by methylene blue dye
reductase test :

Methylene blue dye reductase test :-

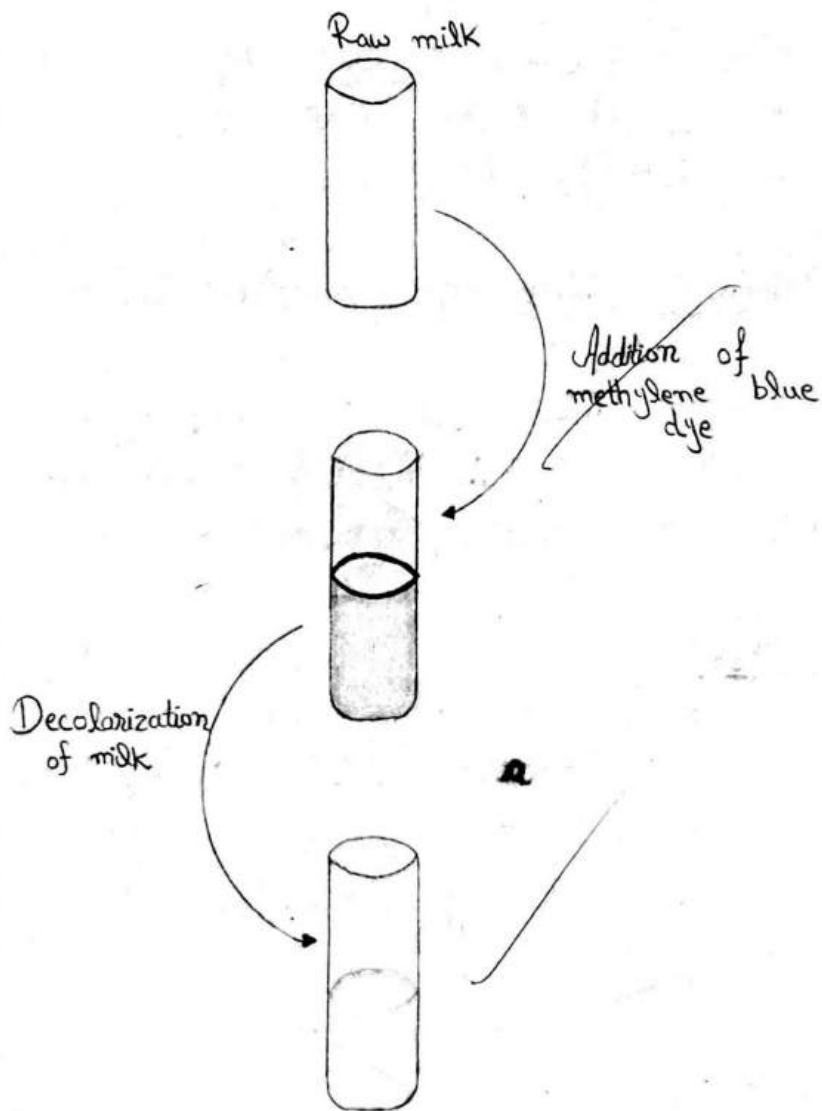
This test is commonly known as MBRT test that is used to check quality of milk.

Principle :-

Methylene blue dye is blue in its oxidation state which is then reduced to colorless compound due to metabolic activities of bacteria in milk.

Apparatus :-

- Test tube
- Test tube stand
- Glass rod
- Pipette



Chemicals:-

- Methylene blue
- Milk sample

Procedure:-

- i- Take test tube and rinse it with water.
- ii- Add 1ml of milk sample in test tube and place it in a test tube stand.
- iii- Now add 1ml methylene blue dye in test tube and mix it well by glass rod.
- iv- Place the test tube containing milk sample in water bath at 37°C and observe it after every 30 minutes.
- v- After two hours, take out the test tube and observe color of milk in it.

Result:-

Milk becomes colorless that means that milk is of good quality.

Experiment No : 6

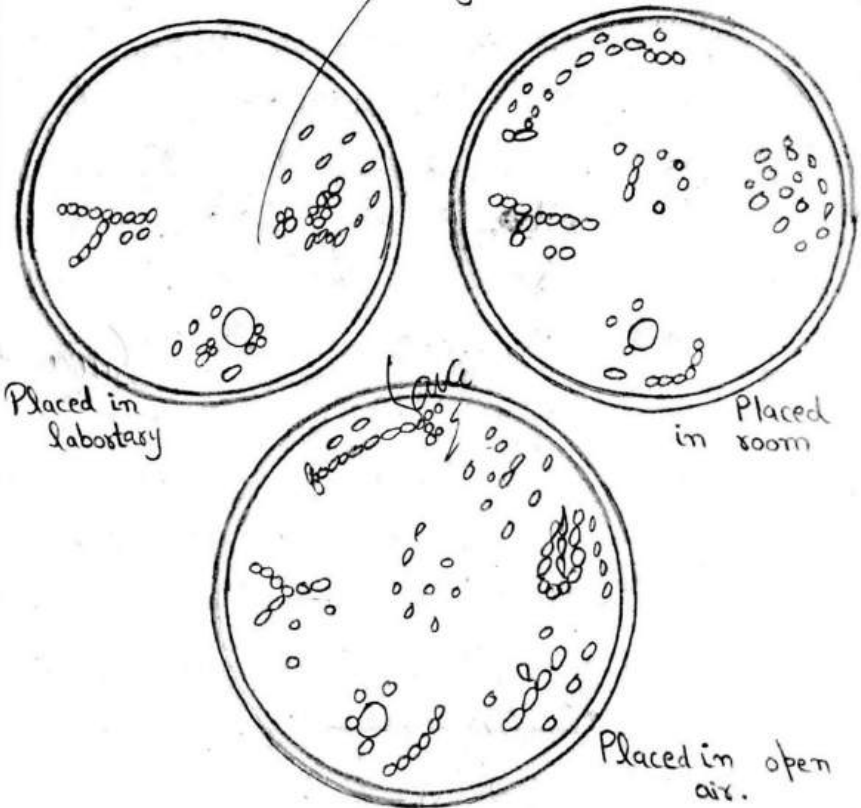
★ Isolation of micro-organisms from air
by expose plate method:

Apparatus:-

- Petri dish
- Laminar flow
- Incubator
- Bunsen burner

Chemicals:-

- Nutrient agar media



Experiment No 6:

Isolation of micro-organisms from air by expose plate method:

Theory:-

Air is not a growth medium, a carrier of particulate matter, dust and droplets which may carry microbes.

Apparatus:-

- Petri dish
- Laminar flow
- Incubator
- Bunsen burner



Chemicals:-

Nutrient agar media

Procedure:-

- i- Pour the liquid nutrient agar media in the petri dishes and allow

Apparatus:-

- Petri dish
- Luminar flow
- Incubator
- Bunsen burner

Chemicals:-

Nutrient agar media

Observation and calculation:-

Sr.No	Name of plates	Average no. of colonies
1	Laboratory	3
2	Room	5
3	Open air	7



to solidify for 5 minutes.

- ii- After solidification expose the plate for 5-10 minutes at different places i.e laboratory room and open air.
- iii- The plates exposed time and date was written on the plates.
- iv- After 5-10 minutes, place the petri plate in the incubator at 37°C for 24-48 minutes.
- v- Observe the plates after 24-48 minutes.

Result:-

Micro-organisms were formed large number of colonies in the petri plate that is exposed in open air which shows that number of microbes vary at different places.

Precautions:-

- i- Exposed time should be correct.

- ii- Incubation time should be fixed.
- iii- Count number of colonies carefully.

Save



Experiment No: 7

★ Isolation of micro-organisms from soil sample by spread plate method:

Apparatus:-

- Beaker
- Pipette
- Spatula
- Weighing balance
- Petri dish
- Laminar flow
- Autoclave

Chemicals:-

- Nutrient agar
- Soil sample
- Water



Experiment No 7:

Isolation of micro-organisms from soil (sample) by spread plate method:

Theory:-

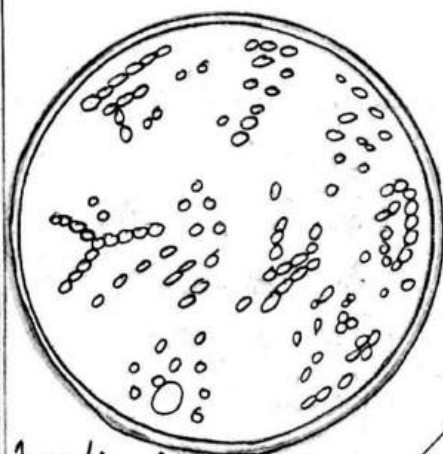
Soil can be defined as the region of earth's crust where geology and biology meet. Soil is the main source of microbes growth.

Apparatus:-

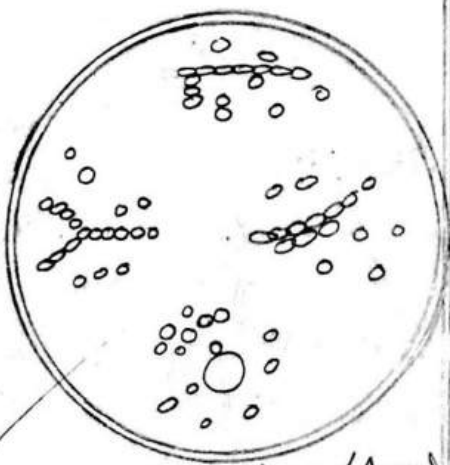
- Beaker
- Pipette
- Spatula
- Weighing balance
- Petri dish
- Laminar flow hood
- Autoclave

Chemicals:-

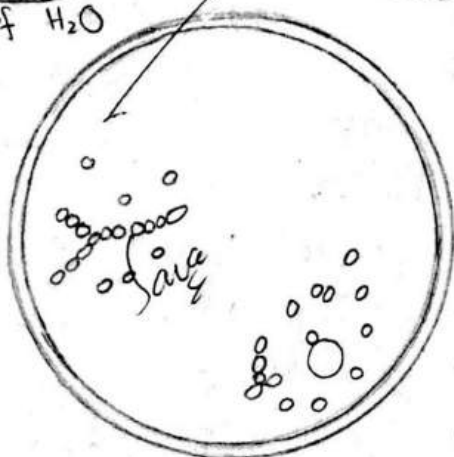
- Nutrient agar medium
- Soil sample



1mg/10ml of H_2O



1mg/100ml of H_2O



1mg/1000ml of H_2O

• Water

Procedure:-

- i- Nutrient agar medium was prepared in petri dish were autoclaved.
- ii- 1gm of soil sample was taken in three different beaker.
- iii- Add 10ml, 100ml and 1000ml water in three beakers respectively.
- iv- Then 1ml soil solution from each beaker put on prepared plate.
- v- Incubate then for 24 hours at room temperature.
- vi- Individuals cells were grow as isolated colonies.

Result:-

- i- After incubation period micro-organisms show colonies which are different in morphology.
- ii- It was also shown that the higher concentration of soil has more

Apparatus:-

- Beaker
- Pipette
- Weighing balance
- Petri dish
- Laminar flow
- Autoclave



Chemical:-

- Nutrient agar
- Soil sample
- Water

Observation and Calculation:-

No. of Plates	Quantity of soil in H_2O	Average no. of colonies
1	1gm/10ml of H_2O	9
2	1gm/100ml of H_2O	4
3	1gm/1000ml of H_2O	2

microbes then the low concentration of soil.

Precautions:-

- i- Incubation time should be fixed.
- ii- Count no. of colonies carefully. San

